

AAROHI GANDHI

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EDUCATION

University of Washington, Seattle Expected Jun 2028

B.S. Applied Mathematics with Computer Science (Paul G. Allen School of CS&E) · GPA 3.80 / 4.00

Admitted to UW at age 15 through the Robinson Center Early Entrance Program, two years ahead of cohort. Dean's List, 4.0 quarter.

Algorithms and Data Structures, Databases and Concurrency, Discrete Math and Logic, Machine Learning, Linear Algebra, Applied Probability, Statistical Inference, Optimization, Combinatorics.

PROJECTS

Adaptive KV Cache Compression for Long Context LLM Inference 2026

Python · PyTorch · Triton · GPU profiling · CUDA memory

- **Cut peak GPU memory 40% at under 2% quality loss** by throwing out the HuggingFace generate abstraction, writing a custom greedy decode loop, and taking manual control of the KV cache memory layout for per head and per layer eviction.
- **Made four eviction methods directly comparable for the first time** by building a single pluggable policy interface and running published baselines (StreamingLLM, H2O, SnapKV) plus a novel adaptive per head budget method under identical instrumentation.
- **Located the decode bottleneck at 30% of per token latency** by profiling the hot loop and instrumenting peak memory, per token decode latency, and attention entropy; evaluated on LongBench, needle in a haystack sweeps, and PG19 perplexity against preregistered criteria.

Screenshot Brain, Semantic Screenshot Search 2026

Next.js · React · Supabase (Postgres, pgvector) · Claude API · embeddings

- **Built a full stack app that makes any screenshot searchable in plain language** by ingesting images, extracting text with a vision model, embedding the content, and serving natural language semantic search over a pgvector index.
- **Made search quality measurable rather than guessed at** by building a labeled evaluation harness that scores top 3 retrieval accuracy; owned the upload pipeline, storage schema, auth, and search UI end to end.

Concurrent Scheduling Backend 2026

Python · SQL · ACID transactions · two phase locking

- **Held transactional correctness across 1,000+ simulated concurrent users** by implementing two phase locking and transaction isolation over a relational schema built from an empty file, with parameterized queries throughout. Cut average query latency over 40% by restructuring joins and indexing hot columns. Scored 75 of 75.

PhishGuard AI, Microsoft Imagine Cup 2024 to 2025

Python · NLP · scikit learn · PostgreSQL · REST APIs

- **Led a 5 person team through the full lifecycle of an ML threat detection system** with fault tolerant fallback handling for model failures and malformed inputs; cut breach rate 45% on test data.

Also: PillPall, iOS medication app shipped solo to the App Store (Swift, SwiftUI) · DubHacks 2024 CareVision, real time multimodal vision system, 2nd of 200+ teams, per frame latency cut 35% · Independent robustness experiments on the WILDS public benchmark across 10+ domains, evaluation consistency improved 40% · OCaml interpreter for a typed lambda calculus, self directed.

EXPERIENCE

Founder and Lead Engineer, CyberMinds 2022 to Present

Production Python platform on Linux · 5,000+ monthly active users across 12 countries · 50+ contributors

- **Scaled from zero to 5,000+ monthly active users across 12 time zones** as sole architect of the entire system, from the request gateway through the model inference layer, with no team, no funding, and no prior codebase.
- **Cut request failure rate from 30% to under 3% through upstream provider outages** by building a fault tolerant multi provider proxy with exponential backoff, per provider rate limiting, idempotent writes, and automatic failover on degradation.
- **Cut dashboard load time over 50%** by profiling the slow request path, restructuring joins, and indexing hot columns; ship on two week cycles with code review and tests on every commit across a 50+ person distributed team.

Founder, Blu Birds 2023 to Present

Technology access initiative for elderly communities · Seattle, WA

- **Delivered usable technology training to 200+ elderly participants** by running structured interviews at retirement communities, publishing a white paper on senior usability barriers, and rebuilding the curriculum around the failure modes they surfaced.

TECHNICAL SKILLS

Languages: Python (primary, expert), SQL (advanced), Java, TypeScript and JavaScript, R, OCaml, Bash

Systems: Linux, concurrency and locking, memory management, performance profiling, latency measurement, ACID transactions, Docker, Git, GitHub Actions, CI/CD, REST APIs, PostgreSQL, SQLite

ML and numerics: PyTorch, Triton, HuggingFace Transformers, CUDA memory profiling, NumPy, pandas, scikit learn, embeddings and vector search, evaluation harness design

Mathematics: Linear algebra, applied probability, statistical inference, discrete math, combinatorics, linear programming and optimization

HONORS

- **Competitive chess player since 2020.** DubHacks 2024, 2nd place of 200+ teams.
- **MIT Beaver Works Summer Institute 2023:** hardware security and embedded systems exploitation.
- Microsoft Imagine Cup competitor · Harvard AI Bootcamp 2024 · Microsoft Certified Cybersecurity Professional · Congressional Gold Volunteer Service Award, 400+ hours.